

KPI	Decrease GHG emissions in line with provincial climate targets
Metric	Estimated GHG emissions avoided from electricity savings (tCO ₂ e)
Calculation	Electricity savings (MWh) × emissions factor (tCO₂e per MWh), where: - Electricity savings = Baseline consumption – Actual consumption - Emissions factor = regional electricity emissions intensity (e.g., BC Hydro grid factor)
Data source	- Electricity consumption data (BC Hydro, Asset Planner) - Building management system (BMS) interval data - Emissions factor from provincial or federal sources (e.g., BC Hydro / Environment Canada)
Explanation of terms/ Scope	This metric translates electricity savings into avoided greenhouse gas (GHG) emissions, enabling alignment between operational energy reductions and broader climate impact. Applies to: - All monitored facilities within FHA's reporting boundary - Scope 2 emissions (electricity-related emissions only) Assumes a consistent emissions factor across reporting periods
Background info	Aligned with Climate Change Accountability targets (50% reduction of emission by public sector buildings by 2030) While BC's electricity grid is relatively low-carbon, reductions in electricity consumption contribute to overall emissions reduction goals.
Data anomalies and trends	- Results depend on emissions factor assumptions - Changes in grid cleanliness affect comparability - External factors (e.g., weather, occupancy, facility expansion) influence electricity consumption and resulting emissions estimates

KPI	
Achieve targeted reduction in electricity consumption across facilities	
Metric	% reduction in electricity consumption compared to baseline
Calculation	$(\text{Baseline electricity consumption} - \text{current consumption}) / \text{Baseline} * 100$
Data source	• Historical baseline data (2007 or define baseline year) • Utility meter data (BC Hydro, Asset Planner) • Building management system (BMS)
Explanation of terms/ Scope	Measures improvement in energy efficiency over time relative to a fixed baseline year. Applies to: • All facilities within FHA's sustainability reporting boundary • Electricity consumption only (Scope 2 emissions driver) Baseline must remain consistent to ensure comparability across reporting periods.
Background info	Aligned with provincial sustainability targets under the Climate Change Accountability Act, which uses a fixed baseline approach for tracking progress Widely used as a primary executive KPI for long-term performance tracking.
Data anomalies and trends	• Influenced by weather variability (heating/cooling demand) • Changes in facility portfolio (new sites, closures) impact comparability • Operational changes (e.g., extended hours) may increase consumption independent of efficiency

KPI	Validate impact of smart transducer implementation
Metric	Electricity savings (MWh) and cost savings (\$) attributable to automation
Calculation	• (Baseline consumption – post-implementation consumption), adjusted for: ○ Weather normalization ○ Occupancy changes ○ Facility operational changes • Cost savings (\$) = Energy saved × applicable electricity rate
Data source	• Utility billing data • BMS / transducer data logs • Project implementation logs
Explanation of terms/ Scope	Measures causal impact of smart transducers, isolating automation-driven savings from external factors. Applies to: • Facilities where transducers are deployed • Defined pre/post implementation periods
Background info	Critical for validating the ROI of smart transducers and scaling decisions across FHA facilities. Supports transition from pilot to enterprise rollout.
Data anomalies and trends	• Attributions may be confounded by concurrent efficiency initiatives • Requires baseline stabilization period • Savings may vary by facility type

KPI	Decrease energy intensity across FHA facilities
Metric	Equivalent kilowatt hours consumed per square metre of facility space (ekWh/m ² /year)
Calculation	(Total electricity consumption (ekWh) divided by gross floor area (m ²) for each facility over the reporting period
Data source	• Utility meter data (BC Hydro, Asset Planner) • Facility floor area data from internal systems (ARCHIBUS)
Explanation of terms/ Scope	Energy intensity normalizes energy consumption across facilities of different sizes to enable comparison. Applies to owned and monitored facilities; comparisons should be made across similar facility types (e.g., hospitals vs outpatient clinics)
Background info	• Used as a standard benchmark metric in energy management across healthcare systems • Supports the identification of inefficient sites and tracking of long-term performance improvements
Data anomalies and trends	Significantly influenced by weather (heating/cooling demand), it may vary due to changes in facility usage or expansion, should not be compared across fundamentally different facility types without segmentation.

KPI	Ensure adoption and effectiveness of automated control systems
Metric	Frequency of manual overrides of automated systems within a reporting period (% or count)
Calculation	Total override events / total system events
Data source	• BMS logs • Control system logs (from actuators)
Explanation of terms/ Scope	Overrides occur when users manually adjust system settings (e.g., lighting, HVAC) that were set automatically by transducers. This metric indicates: • User trust in automation • Alignment of system settings with real operational needs
Background info	High override frequency may indicate: • User discomfort (temperature, lighting levels) • Poor calibration of control algorithms • Misaligned schedules with clinical workflows
Data anomalies and trends	• Higher overrides expected during rollout phase • May vary significantly across departments (clinical vs admin)

KPI	Ensure reliability of smart transducer system
Metric	Percentage of sensors not functioning as expected
Calculation	$(\text{Number of failed sensors} / \text{total sensors}) \times 100$
Data source	• Sensor diagnostics • BMS system logs
Explanation of terms/ Scope	Measures system reliability and data integrity, which are critical for accurate automation and reporting. Applies to all deployed smart transducers and sensors.
Background info	Reliable sensor data is foundational to: • Automation effectiveness • Trust in system outputs • Decision-making accuracy
Data anomalies and trends	• Temporary spikes during maintenance or upgrades • Failure rates may vary by sensor type or environment